



Insight into self-discharge of layered lithium-rich oxide cathode in carbonate-based electrolytes with and without additive



Jianhui Li ^a, Lidan Xing ^{a,b}, Liping Zhang ^c, Le Yu ^c, Weizhen Fan ^c, Mengqing Xu ^{a,b},
Weishan Li ^{a,b,*}

^a School of Chemistry and Environment, South China Normal University, Guangzhou 510006, China

^b Engineering Research Center of MTEES (Ministry of Education), Research Center of BMET (Guangdong Province), and Key Laboratory of ETESPG (GHEI), South China Normal University, Guangzhou 510006, China

^c Guangzhou Tinci Material Technology Co., Ltd, Guangzhou 510760, China

HIGHLIGHTS

- Self-discharge of charged layered lithium-rich oxide cathode is understood.
- Trimethyl borate (TMB) is used as an electrolyte additive to suppress this self-discharge.
- Suppression mechanism involves a cathode film derived from the preferential oxidation of TMB.

ARTICLE INFO

Article history:

Received 4 February 2016

Received in revised form

13 May 2016

Accepted 18 May 2016

Keywords:

Layered lithium-rich oxide

Self-discharge

Electrolyte additive

Cathode film

ABSTRACT

Self-discharge behavior of layered lithium-rich oxide as cathode of lithium ion battery in a carbonated-based electrolyte is understood, and a simple boron-containing compound, trimethyl borate (TMB), is used as an electrolyte additive to suppress this self-discharge. It is found that layered lithium-rich oxide charged under 4.8 V in additive-free electrolyte suffers severe self-discharge and TMB is an effective electrolyte additive for self-discharge suppression. Physical characterizations from XRD, SEM, TEM, XPS and ICP-MS demonstrate that the crystal structure of the layered lithium-rich oxide collapses due to the chemical interaction between the charged oxide and electrolyte. When TMB is applied, the structural integrity of the oxide is maintained due to the protective cathode film generated from the preferential oxidation of TMB.

© 2016 Published by Elsevier B.V.

1. Introduction

Lithium ion battery has been widely used as power source for electronic devices due to its high energy density and long cycle life compared with other rechargeable batteries [1–4]. However, the energy density of currently commercialized lithium ion battery cannot meet the demand of electric vehicles, because the specific capacity of cathode materials such as LiCoO_2 , LiMn_2O_4 and LiFePO_4 is low compared to that of graphite anode [5–7]. Layered lithium-rich oxide, composed of Li_2MnO_3 and LiMO_2 ($\text{M} = \text{Mn}_{1-x-y}\text{Co}_x\text{Ni}_y$), has drawn much attention recently and are considered as a promising cathode for high energy density lithium

ion battery, because it can provide a specific capacity of as high as 300 mAh g^{-1} under high voltage (4.8 V, vs. Li/Li^+) [8–12]. In addition, this oxide is cheaper than LiCoO_2 , the most widely used cathode for lithium ion battery up to date, due to the abundance of manganese.

However, there remain challenges before layered lithium-rich oxide can be put into practical use. To achieve its larger specific capacity, layered lithium-rich oxide cathode needs to be charged over 4.5 V for the activation of Li_2MnO_3 [13–18]. Under such high voltage, the electrolyte tends to be oxidized electrochemically on cathode [19,20], which is accompanied with the crystal destruction of layered lithium-rich oxide, leading to the poor cyclic stability [21–29]. Besides, like other high voltage cathode materials such as $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ [3,30,31], charged layered lithium-rich oxide might fail due to self-discharge when it is kept in storage. The failure mechanism of the cathode due to the self-discharge is different

* Corresponding author. School of Chemistry and Environment, South China Normal University, Guangzhou 510006, China.

E-mail address: liwsh@scnu.edu.cn (W. Li).

from that resulting from charge/discharge cycling, because self-discharge takes place through the chemical interaction between charged cathode and electrolyte [3]. Unfortunately, less attention has been paid to the mechanism on and the solution to the self-discharge of layered lithium-rich oxide.

Surface coating is proven to be an effective way to improve interfacial stability of layered lithium-rich oxide. Many inorganic compounds, such as Al_2O_3 [32], MgO [33], MnO_x [34], AlPO_4 [35,36], AlF_3 [37,38], $\text{Li}_4\text{Ti}_5\text{O}_{12}$ [39], or their composites [40,41], have been used to coat layered lithium-rich oxide. However, coating is not cost-efficient because coating procedure is complicated and introducing inert compounds reduces the specific capacity of layered lithium-rich oxides.

Compared to the coating, using electrolyte additives to modify the electrode-electrolyte interface is easy and economical. Some electrolyte additives have been developed successfully to improve the cyclic stability of layered lithium-rich oxide. These additives include 4-(trifluoromethyl)-1,3-dioxolan-2-one (TFM-EC) [42], tris(hexafluoro-iso-propyl)phosphate (HFip) [43], 3-Hexylthiophene [44], LiDFOB [45], tris(trimethylsilyl)phosphate (TMSP) [46], tris(trimethylsilyl)borate (TMSB) [47]. This cyclic stability improvement is attributed to the formation of cathode film on layered lithium-rich oxide, which suppresses the electrolyte decomposition and prevents the oxide from structural destruction during charge/discharge cycling. The cathode film should also be beneficial for the self-discharge suppression of layered lithium-rich oxide, because it provides a separation layer and avoids the direct contact between the oxide and electrolyte.

In our previous report, a simple and cheap boron-containing compound, trimethyl borate (TMB), was found to be effective for the cyclic stability improvement of a cathode, $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$, under 4.5 V [48]. This improvement is attributed to the formation of a protective cathode film on $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ particles from preferential oxidation of TMB, which not only suppress the electrolyte decomposition, but also protects cathode from crystal destruction. Boron in the additive is believed to be incorporated into the cathode film, which coordinates with F ions preventing the formation of LiF on one hand, and contributes to the improved ionic conductivity of the resulting films and the improved rate capability of the protected cathodes on the other hand [49–52]. In this work, the self-discharge mechanism of layered lithium-rich oxide in carbonate-based electrolyte was understood and TMB was used as an electrolyte additive to suppress the self-discharge of charged layered lithium-rich oxide.

2. Experimental

2.1. Theoretical calculation

The energy levels of highest occupied molecular orbital (HOMO) of organic molecules were calculated by the Gaussian 03 program package [11], for the comparison in oxidative possibility of TMB with solvents, ethylene carbonate, ethyl methyl carbonate and diethyl carbonate.

2.2. Sample preparations

A representative layered lithium-rich oxide, $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$, was used in this work. It was prepared by co-precipitation method [53]. Typically, 1 g polyvinyl pyrrolidone was dispersed under magnetic stirring in the mixed solvent of 60 mL distilled water and 20 mL ethylene glycol. 4.875 g lithium acetate ($\text{CH}_3\text{COOLi} \cdot 2\text{H}_2\text{O}$), 4.901 g manganese acetate ($\text{Mn}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$), 1.224 g nickel acetate ($\text{Ni}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$), and 1.245 g cobalt acetate ($\text{Co}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$) were added

successively under continuous stirring. An oxalic acid solution (120 mL distilled water and 9.331 g oxalic acid) as a precipitating agent was added drop by drop into the solution above under stirring. After complete precipitation, the mixture was evaporated at 90 °C. Finally, the dried powder was pre-heated in air at 450 °C for 5 h and then calcined in air at 900 °C for 12 h. The final product was obtained after uniform grinding.

TMB (>99.5%) was purchased from Aladdin Industrial Inc., China, and used without further purification. Battery-grade carbonate solvents and lithium hexafluorophosphate (LiPF_6) were provided by Dongguan Kaixin Materials Technology Co. Ltd, China. The baseline electrolyte was 1.0 M LiPF_6 in a mixed solvent of ethylene carbonate (EC)/ethyl methyl carbonate (EMC)/diethyl carbonate (DEC) (3/5/2, in weight). 2 wt% TMB was added to obtain the TMB-containing electrolyte. All the electrolytes were prepared in a highly pure argon-filled glove box (MBRAUN, Germany), in which water and oxygen contents were controlled less than 0.1 ppm. The contents of water and free acid (HF) in the electrolyte were determined by Karl-Fisher 831 Coulometer (Metrohm, Switzerland) and Karl-Fisher 798 GPT Titrino (Metrohm, Switzerland), which were controlled lower than 20 and 50 ppm, respectively.

The cathode was prepared with a slurry of 80 wt% of as-prepared $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$ powder, 10 wt% polyvinylidene fluoride (PVDF) binder and 10 wt% acetylene black. The slurry was cast on aluminum foil, followed by drying and calendaring. The electrode area was 1.2 cm^2 and its thickness was 28 μm . $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$ 2025-coin cell was assembled with Celgard 2400 separator in the glove box. Transparent V-type cells with lithium as anode and $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$ as cathode were also assembled for phenomenological observation.

2.3. Electrochemical measurements

Cyclic voltammetry (CV) was performed on a Solartron-1470 instrument (U.K) at a scan rate of 0.1 mV s^{-1} in the coin cell. Self-discharge test of coin cells was performed according to the following protocols. The cells were charged and discharged at 0.1 C for the first 3 cycles and 0.5 C for additional 17 cycles (1 C = 250 mA g^{-1}) in the voltage range of 2–4.8 V with the end-of-voltage of 4.8 V on LAND test system (LandCT2001A, China) and then left at open-circuit voltage for 18 days. The cycling performance of $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$ coin cells after self-discharge test was re-evaluated at 0.5 C rate in the voltage range of 2–4.8 V. Electrochemical impedance spectroscopy was performed on a PGSTAT-30 electrochemical station (Autolab, Metrohm, Netherlands) in a frequency range of 10^5 –0.1 Hz with a potential amplitude of 5 mV.

2.4. Physical characterizations

The cycled $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.13}\text{Mn}_{0.54}\text{Co}_{0.13}]\text{O}_2$ electrodes were rinsed with anhydrous DMC three times to remove residual electrolyte salts precipitated on the surface, and then evacuated overnight at room temperature for physical characterizations [54]. Crystal structure was identified by X-ray diffraction (XRD, BRUKER D8 ADVANCE, Germany) operated at 60 kV and 80 mA using $\text{CuK}\alpha$ radiation in the 2θ range of 15–90° with a sweep rate of 0.05°s^{-1} . Surface morphology was observed by scanning electron microscopy (SEM, JSM-6510) and transmission electron microscopy (TEM, JEM-2100, JOEL, Japan). Surface compositions were determined by X-ray photoelectron spectroscopy (XPS, ESCALAB 250), using a focused monochromatized Al $\text{K}\alpha$ radiation ($h\nu = 1486.6 \text{ eV}$) under ultra-high vacuum. The obtained spectra were fitted using XPS peak software (version 4.1). Lorentzian and Gaussian functions

were used for the least-square curve fitting procedure. The organic functional groups in cathode films were identified by Fourier transform infrared spectroscopy (FTIR, Bruker Tensor 27) within 600–1800 cm^{-1} . To determine the contents of transition metal ions deposited on lithium electrode, the lithium electrodes taken from the cycled cells were rinsed with anhydrous DMC and dissolved in 10 mL 2% HNO_3 , which then was diluted to 25 mL. The contents of Mn, Ni and Co in the cycled lithium electrodes and electrolyte of V-type cell were analyzed by inductively coupled plasma atomic emission spectrometer for Inductively Coupled Plasma (ICP) analysis, which was performed on an IRIS Intrepid II XSP.

3. Results and discussion

3.1. Self-discharge of charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$

The self-discharge of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ was evaluated by observing the variation of cell voltage after $\text{Li}/\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ coin cell was performed with charge/discharge cycling ending at 4.8 V Fig. 1 shows the voltage variation during self-discharge test and the cycling performance after self-discharge test for $\text{Li}/\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ coin cell in baseline and TMB-containing electrolytes. As seen from Fig. 1A, after storage for 18 days, the voltage of $\text{Li}/\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ coin cell in baseline electrolyte decreases from 4.8 V to 1.0 V, while the cell in TMB-containing electrolyte remains almost unchanged. It is apparent that the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ in baseline electrolyte suffers a serious self-discharge, which can be suppressed effectively by applying TMB.

In baseline electrolyte, the carbonate electrolytes tend to lose electrons and oxidatively decompose on the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$, corresponding to the potential plateaus at 4.2–4.5 V from point a to b [12,55,56]. Accordingly, the transition metal ions with high valence such as Ni^{4+} , Co^{4+} and Mn^{4+} on the surface of the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ are reduced to some extent. Since a protective cathode film cannot be formed from the electrolyte decomposition products, the electrolyte oxidation decomposition successively proceeds and most of surface transition metal ions are reduced into those with lower valence such as Ni^{2+} or Ni^{3+} , Co^{3+} and Mn^{3+} . These reductions corresponds to the potential drop from b (4.2 V) to c (3.0 V), which is accompanied with lithium insertion into the layered structure and do not involve phase transition [31,57]. The oxidation decomposition of the electrolyte and the reduction of transition metal ions result from the chemical interaction between electrode material and electrolyte and take place simultaneously. These reactions are different from those during charge/discharge cycling in which the reductions of

transition metal ions take place during discharge process while the oxidation decomposition of electrolyte takes place during charge process [3]. In the case of self-discharge, HF generated from electrolyte decomposition might cause the crystal destruction of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ cathode, forming new compounds with fully inserted lithium. The new compounds increase the volume of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ particles [31] and isolate $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ particles from current collector or carbon conductor, leading to the exposure of current collector (aluminum) to the electrolyte and the dissolution of aluminum oxide on the current collector. The formation of new compounds and the exposure of current collector account for the abrupt potential change from point c to d, while the dissolution of aluminum oxide on the current collector is responsible for the potential change from the point d to e. From the point e to f, the potential remains at about 1 V, indicative of the exposure of metal aluminum.

In the TMB-containing electrolyte, a protective cathode film can be formed on $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ surface due to the preferential oxidation of TMB [47], which separates $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from direct contact with electrolyte and thus suppresses the self-discharge of the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$. The crystal destruction of the cathode and the protection that TMB yields can be confirmed by comparing the cycling performance of the cells after self-discharge test.

Fig. 1B shows the cycling performance of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode in baseline and TMB-containing electrolytes before and after self-discharge test. Before the self-discharge test, the electrode in the TMB-containing electrolyte exhibits better cyclic stability than the electrode in the baseline electrolyte, suggesting that TMB is also beneficial to the cyclic stability improvement of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$. After self-discharge test, the electrode in the baseline electrolyte shows a significant capacity loss, from 194 to 159 mAh g^{-1} . This capacity loss suggests that partial of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ suffers crystal destruction, which can be explained by the self-discharge of surface $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ that directly contacts with electrolyte. Comparatively, the discharge capacity of electrode in the TMB-containing electrolyte changes slightly after self-discharge test, indicating that TMB provides a protection for $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from crystal destruction. Nevertheless, a minor capacity loss still exists for the electrode in TMB-containing electrolyte, which is inevitable when the charged electrode is kept in storage for 18 days. The far lower coulombic efficiency of the electrode in the baseline electrolyte than that in TMB-containing electrolyte, as shown in Fig. 1C, suggests that destructed $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ accelerates the electrolyte oxidation decomposition during charge process.

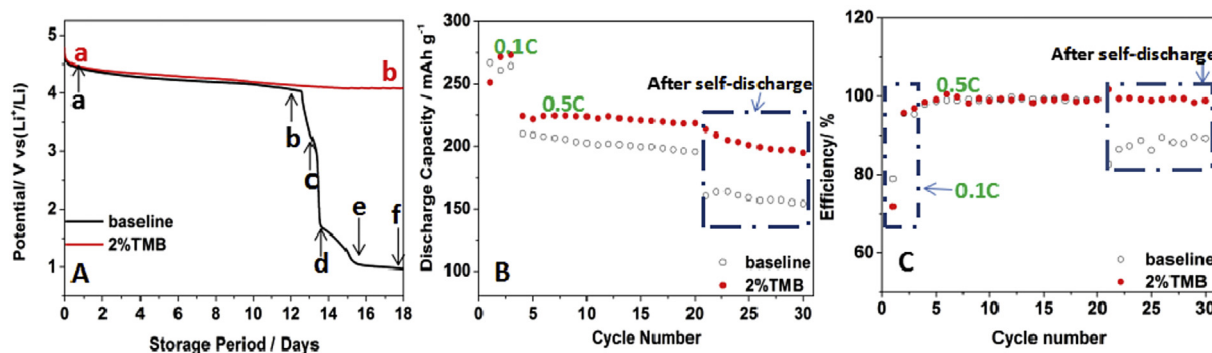


Fig. 1. V-t (A) curves during self-discharge, cycling performance (B) and coulombic efficiency (C) of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode before and after self-discharge in baseline and TMB-containing electrolytes.

3.2. Contribution of TMB

The self-discharge suppression of the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ is ascribed to the cathode film generated from TMB, which separates $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from the direct contact with electrolyte. To confirm this contribution, further electrochemical analyses and physical characterizations were conducted.

A protective film on cathode from an electrolyte additive should be formed before the oxidation decomposition of solvents in electrolyte, because gas evolution generated from electrolyte decomposition interferes with the formation of cathode film [49]. The oxidation activity of TMB was compared with the solvents in baseline electrolyte by DFT calculation. The obtained results are presented in Table 1. It has been known that the highest occupied molecular orbital (HOMO) energy corresponds to the oxidation preference of organic molecules. The higher the HOMO level of a molecule is, the easier the molecule is oxidized. It can be seen from Table 1 that TMB has a higher HOMO energy than DEC, EC and EMC, suggesting that TMB might be oxidized preferentially to the baseline electrolyte. This preference can be identified by comparing the initial charge/discharge profiles and corresponding differential charge capacity (dQ/dV) curves, as shown in Fig. 2. Fig. 2A shows the initial charge/discharge profiles for $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode in the baseline and TMB-containing electrolytes. The electrode in the TMB-containing electrolyte has a slightly lower initial coulombic efficiency (67%) than the electrode the baseline electrolyte (73%), indicating that the oxidation decomposition of TMB happens during charging process. The dQ/dV curves in Fig. 2B show that TMB oxidation takes place at 3.79 V, which is lower than the potential for the initial lithium extraction from $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ (3.91 V). Obviously, TMB is oxidized preferentially to the carbonate-based electrolyte that is stable at the potential lower than 4.0 V [11,12]. Fig. 2C presents the cyclic voltammograms of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ in the baseline and TMB-containing electrolytes. There are two oxidation peaks centered at ~4.0 and ~4.7 V, corresponding to the oxidation of the transition metals ($\text{Ni}^{2+} \rightarrow \text{Ni}^{3+} \rightarrow \text{Ni}^{4+}$ and $\text{Co}^{3+} \rightarrow \text{Co}^{3.6+}$) [58], and oxygen loss with Li extraction ($\text{Li}_2\text{MnO}_3 \rightarrow 2\text{Li}^+ + 2\text{e}^- + 0.5\text{O}_2 + \text{MnO}_2$) [59], respectively. Correspondingly, there are three reduction peaks distinguishable at ~4.3, ~3.7 and ~3.3 V. The reduction peaks at ~4.3 and 3.7 V are apparently due to the reduction of Ni^{4+} and $\text{Co}^{3.6+}$, while the peak at ~3.3 V could be ascribed to the reduction of Mn ions ($\text{Mn}^{4+} \rightarrow \text{Mn}^{(4-x)+}$ ($0 < x < 1$)) [60]. It can be seen from Fig. 2C that the oxidation current between 3.7 and 4.0 V for the electrode in the TMB-containing electrolyte is larger than that in the baseline electrolyte, confirming the preferential oxidation of TMB on the electrode. Therefore, the cathode film generated from TMB avoids the interference from electrolyte decomposition products and yields an effective separation of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from electrolyte.

With a protective cathode film available, it can be expected that the electrolyte decomposition can be suppressed and the structure integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ can be maintained. The suppression of electrolyte decomposition by applying TMB can be confirmed by the chronoamperometric responses in Fig. 3A, which were obtained at 5.0 V for the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$

electrodes in baseline and TMB-containing electrolytes after 20 charge/discharge cycles (0.1 C for first three cycles and then at 0.5 C for the subsequent cycles). The residual current, representing the electrolyte decomposition, is 0.1 mA cm^{-2} for the electrode in baseline electrolyte, but can be hardly recorded for the electrode in TMB-containing electrolyte. Obviously, the application of TMB suppresses effectively the electrolyte decomposition.

The contribution of TMB to the structure integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ can be indicated by the change in impedance spectra before and after self-discharge test, as shown in Fig. 3B. The impedance spectra are characteristic of a semicircle at high frequencies and a slope line at low frequencies. The semicircle at high frequencies reflects the interfacial impedance including cathode film and charge-transfer resistances, while the slope line at low frequencies is corresponding to the Warburg impedance [61]. After storage, the impedance spectrum of the electrode in baseline electrolyte changes significantly, suggesting that the electrode has suffered structural destruction. Differently, the impedance spectrum of the electrode in TMB-containing electrolyte is hardly affected by the storage for self-discharge test, suggesting that the structure integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ can be maintained by applying TMB.

The structure integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ can be also identified by the change in discharge profiles before and after self-discharge test. Fig. 4 presents the discharge profiles and corresponding dQ/dV curves of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes in baseline and TMB-containing electrolytes before and after self-discharge test and Table S1 presents a comparison in discharge capacity under different potentials. After self-discharge test, as seen from Fig. 4A, there is a significant change in the discharge capacity and potential plateau for the electrode in baseline electrolyte, while the discharge profile remains almost unchanged for the electrode in TMB-containing electrolyte. The discharge capacity of layered lithium-rich oxide include two lithiation processes: one is that at the potential above 3.5 V, corresponding to the redox couples of $\text{Ni}^{4+}/\text{Ni}^{2+}$ and $\text{Co}^{3.6+}/\text{Co}^{3+}$, the other is that at the potential below 3.5 V, corresponding to the redox couple of $\text{Mn}^{4+}/\text{Mn}^{(4-x)+}$ ($0 < x < 1$) [30]. As shown in Table S1, both electrodes exhibit various discharge capacity proportion for these lithiation processes, which has a significant change for the electrode in baseline electrolyte but remains almost unchanged for that in TMB-containing electrolyte. This comparison suggests that crystal structure of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ collapses in the baseline electrolyte but can be protected effectively by applying TMB. As shown in Fig. 4B, three main reduction processes can be identified in dQ/dV curves: Re1 around 4.5 V is related to Li occupation in octahedral sites, Re2 at 3.7 V can be ascribed to Li occupation in octahedral sites associated with the $\text{Ni}^{4+}/\text{Ni}^{2+}$ and $\text{Co}^{4+}/\text{Co}^{3+}$ and Re3 at lower than 3.5 V can be explained by Li occupation in octahedral sites corresponding to $\text{Mn}^{4+}/\text{Mn}^{3+}$ redox [47]. It can be found from Fig. 4B that the peak at 3.7 V is severely weakened in the baseline electrolyte after self-discharge test. Differently, three main reduction peaks in TMB-containing electrolyte remains almost unchanged. The appearance of Ni^{4+} could be accompanied with the transformation into a Li-rich spinel $\text{Li}(\text{Li}_{1/3}\text{Ni}_{1/2}\text{Mn}_{7/6})\text{O}_4$, which takes place mainly on the surface region as a result of the attack from HF [60,62]. Since Ni^{4+} in spinel $\text{Li}(\text{Li}_{1/3}\text{Ni}_{1/2}\text{Mn}_{7/6})\text{O}_4$ does not contribute to the discharge capacity, the 3.7 V discharge potential plateau disappears, which explains the capacity loss of the electrode in the baseline electrolyte after self-discharge test. The application of TMB suppresses the formation of HF and protects $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from structural destruction.

Transparent V-type cells with lithium as anode and $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ as cathode were assembled for observing the phenomenological change due to the self-discharge. Fig. 5

Table 1
HOMO energies of DEC, EC, EMC and TMB.

Molecules	HOMO energy (eV)
Dimethyl carbonate (DEC)	−0.30694
Ethylene carbonate (EC)	−0.32081
Ethyl methyl carbonate (EMC)	−0.30867
Triethyl borate(TMB)	−0.28995

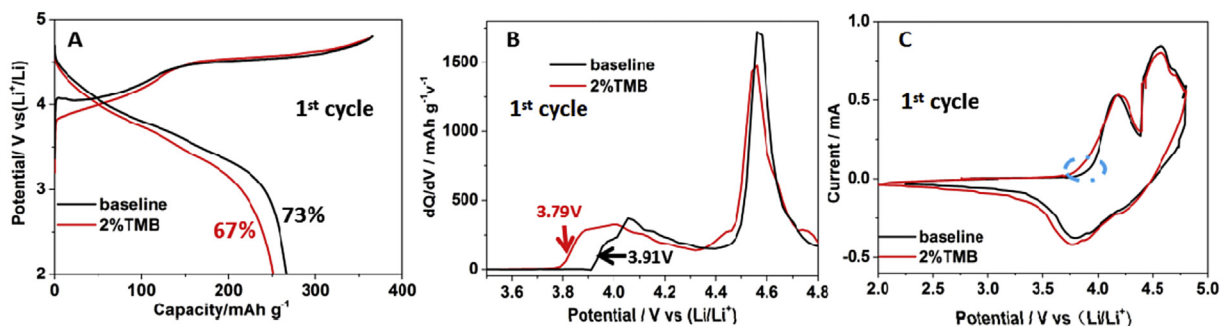


Fig. 2. Initial charge/discharge profiles (A), differential charge capacity curves (B) and cyclic voltammograms (C) of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode in baseline and TMB-containing electrolytes.

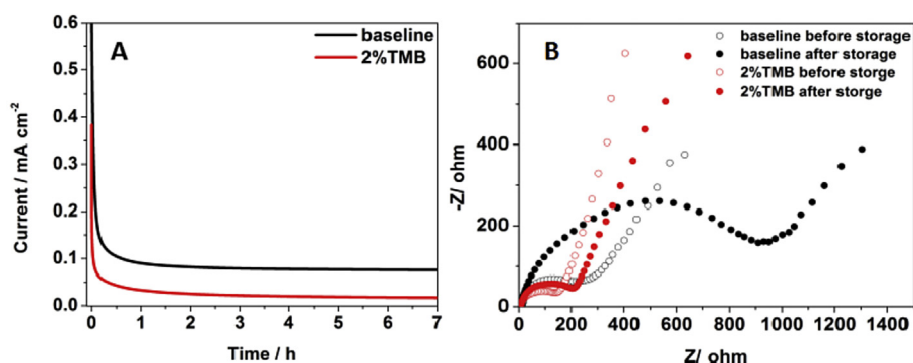


Fig. 3. Chronoamperometric responses (A) and electrochemical impedance spectra (B) of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode in baseline and TMB-containing electrolytes.

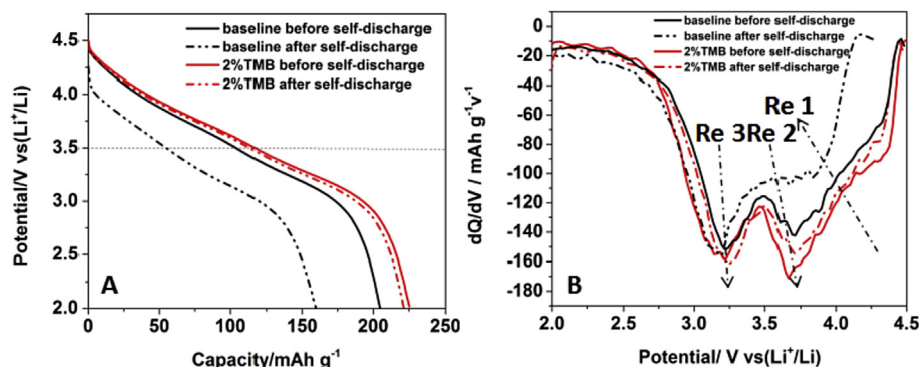


Fig. 4. Discharge profiles (A) and corresponding differential capacity curves (B) of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode before and after self-discharge in baseline and TMB-containing electrolytes.

presents the photos of V-type cell before and after self-discharge test, which was performed with same protocol as did on coin cell. The electrolytes in both V-type cells show color change after self-discharge test, which is attributed to the decomposed electrolyte products and the dissolved transition metal ions from cathode due to the chemical interaction between the charged cathode and electrolyte [3,11,12]. The color is far deeper for the cell in the baseline electrolyte than the cell in the TMB-containing electrolyte, indicating that the self-discharge of the charged cathode take place seriously in the baseline electrolyte but can be suppressed to a great extent by applying TMB. As shown in Table 2, the far larger contents of transition metal ions in baseline electrolyte than in TMB-containing electrolyte suggest that $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ suffers structural destruction in the baseline electrolyte and the application of TMB yields an effective protection.

3.3. Cathode film generated from TMB

All the contributions mentioned above result from the cathode film generated from the preferential oxidation. To identify the existence of the cathode film on $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$, surface morphology and composition of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ particles after self-discharge test were compared with pristine one by SEM, TEM, XPS and FTIR.

Fig. 6 presents the SEM and TEM images of the fresh $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode and those after self-discharge test. It can be seen from Fig. 6A and D that the surface of the fresh electrode is clean and smooth. However, after self-discharge test in the baseline electrolyte, as shown in Fig. 6B and E, the electrode surface is apparently covered with thick and uneven deposits. Additionally, the well-crystallized particle (Fig. 6D) turns

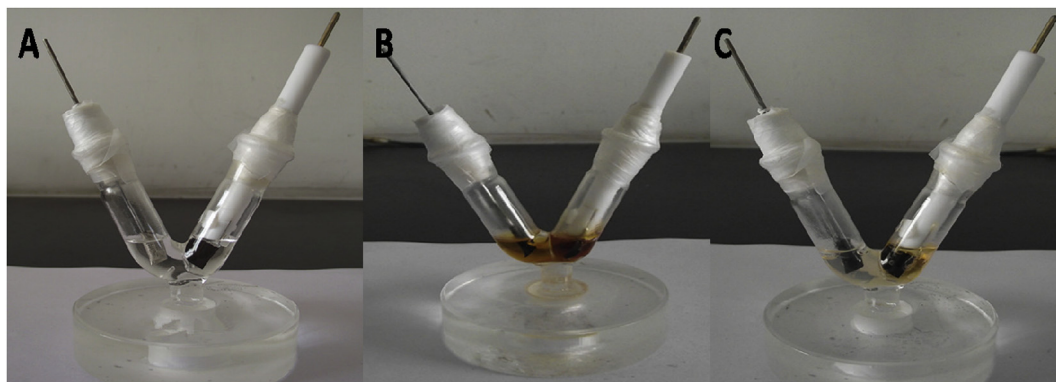


Fig. 5. Photograph of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ V-type cells: fresh (A), after self-discharge in baseline (B) and TMB-containing (C) electrolytes.

Table 2

Contents of transition metal ions (in ppm) in the electrolytes of V-type cells.

	Mn	Ni	Co
Baseline	6.07	15.06	1.21
TMB	2.67	3.14	0.67

into pulverization (Fig. 6E). The thick deposits should be ascribed to the products from the successive decomposition of the electrolyte, while the pulverization indicates that the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ suffers structural destruction [63]. Differently, the surface morphology of the electrode in TMB-containing electrolyte maintains the crystal integrity and a thin and uniform (with a thickness of about 10 nm) film can be clearly identified, as shown in Fig. 6F. This observation confirms that cathode film has been formed on $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ particles due to the preferential oxidation of TMB. This cathode film separates $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from direct contact with electrolyte and thus suppresses the self-discharge of the charged $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ cathode. With this cathode film, the electrolyte decomposition can be inhibited effectively and the crystal integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ can be well maintained.

Fig. 7 presents the X-ray diffraction patterns of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes after self-discharge test, with a comparison of pristine electrode. It can be found from Fig. 7 that the pristine electrode exhibits a typical diffraction pattern of layered structure, together with that of aluminum current collector [53]. For the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ in the baseline electrolyte, all the characteristic peaks of layered structure are weakened and even disappear. However, the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ in the TMB-containing electrolyte retains same diffraction pattern as pristine one. This characterization indicates that the structure integrity of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ collapses when it suffers self-discharge and the cathode film formed from TMB protects $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ from structural destruction.

The surface composition of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ were analyzed by XPS. Fig. 8 presents the obtained XPS spectra of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrode after self-discharge test. The C1s spectra show that PVDF binder (290.3 eV and 285.7 eV) and graphite (284.8 eV) are presented in the fresh and cycled cathodes. The weaker peaks for C–C and PVDF of the cycled cathode in the baseline electrolyte than that in TMB-containing electrolyte indicate that graphite and PVDF is more difficult to detect on the former than the later. This difference can be explained by the thick deposits

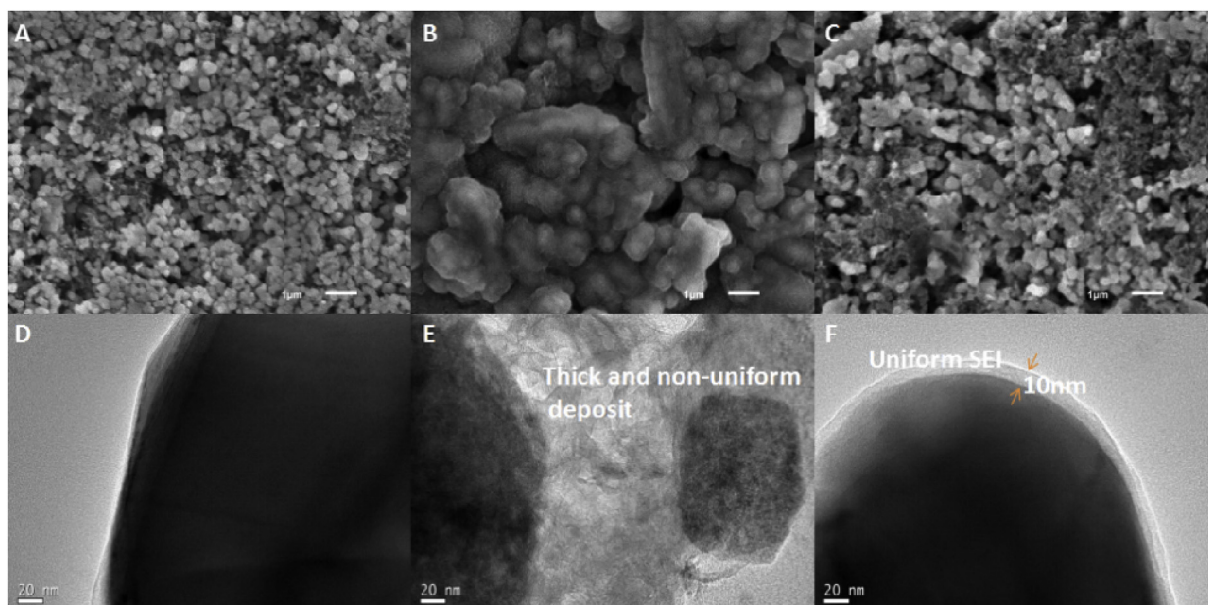


Fig. 6. SEM and TEM images of fresh (A and D) and the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes after self-discharge in baseline (B and E) and TMB-containing (C and F) electrolytes.

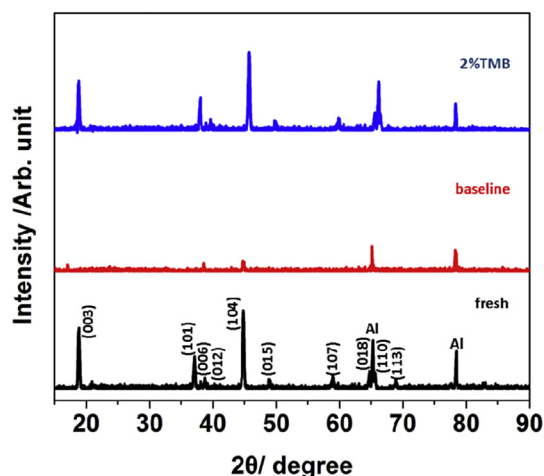


Fig. 7. XRD patterns of fresh and the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes after self-discharge in baseline and TMB-containing electrolytes.

on the former and the thin film on the latter. The peak intensities of C=O (289 eV) and C–O (286 eV), corresponding to the electrolyte decomposition products, ROCO_2Li , ROLi and Li_2CO_3 [64], are stronger for the electrode cycled in baseline electrolyte than that in TMB-containing electrolyte, indicative of the serious decomposition of the baseline electrolyte. In the O 1s spectra, the Me–O (529.9 eV) corresponds to the bonds between transition metal and oxygen, while the C=O (531.7 eV) and C–O (533.4 eV) to the polycarbonates formed from electrolyte decomposition [65–70]. Compared with the electrode in baseline electrolyte, the Me–O peak of the electrode in TMB-containing electrolyte becomes

stronger, while the C=O peak become weaker. Similarly in the F 1s and P 2p spectra, the LiF and $\text{Li}_x\text{PO}_y\text{F}_z$ peaks are weaker in baseline electrolyte than in TMB-containing electrolyte. These differences indicate that electrolyte decomposition products on the electrode cycled in TMB-containing electrolyte are less than those in baseline electrolyte. Mn and Ni can also be identified for both cycled electrodes in the baseline and TMB-containing electrolytes, which are from layered lithium-rich oxide. In the Ni 2p spectra, there are main peaks for Ni $2p_{3/2}$ (Ni^{4+}) and Ni $2p_{1/2}$ are 853 and 868 eV, and two satellite peaks for Ni^{2+} at 858 and 874 eV [71]. Compared with the fresh electrode, the Mn and Ni peaks becomes weaker for the electrode cycled in baseline electrolyte but remains almost unchanged for the electrode cycled in TMB-containing electrolyte, confirming that the crystal structure of layered lithium-rich oxide collapses in the baseline electrolyte but is maintained in TMB-containing electrolyte.

The peak of B–O in the B 1s spectrum can be identified for the electrode in the TMB-containing electrolyte but not in the baseline electrolyte, confirming that the cathode film is generated from the oxidation decomposition of TMB. It has been known that the composition of cathode film might be affected by rinsing [54]. To confirm whether boron-containing film is affected by the rinsing of the electrode, XPS was performed on the cycled electrodes with and without rinsing. The obtained results are presented in Fig. S1, which shows that both electrodes exhibit similar XPS pattern, indicating that the boron-containing film is not affected by rinsing. The boron-containing film can also be confirmed by FTIR spectra, as shown in Fig. S2. The IR spectra are dominated by the absorptions characteristics of PVDF (1400, 1271, 1170, 1070, 877 and 727 cm^{-1}) for both electrodes in baseline and TMB-containing electrolyte. Differently from the electrode in baseline electrolyte, the electrode in TMB-containing electrolyte shows additional peaks at

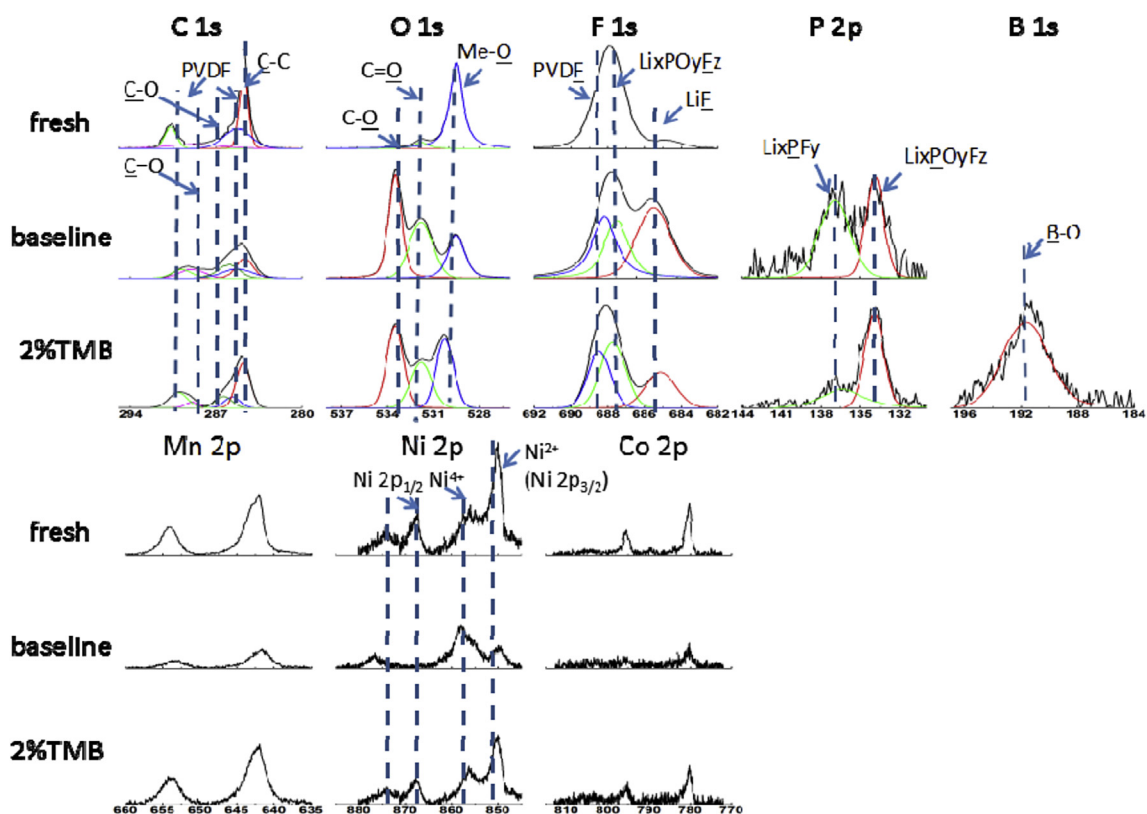


Fig. 8. XPS profiles of fresh and the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes after self-discharge in baseline and TMB-containing electrolytes.

Table 3

Contents of transition metal (in ppm) on lithium anodes in baseline and TMB-containing electrolytes after self-discharge test.

	Mn	Ni	Co
Baseline	7.08	18.34	1.58
TMB	3.22	5.26	1.02

1635 cm^{-1} , which are characteristics of B–O species [3]. The preferential oxidation of TMB forms a protective cathode film, which suppresses the electrolyte decomposition and maintains the structure stability of layered lithium-rich oxide.

The structural destruction of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ is accompanied by the dissolution of transition metal ions from $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$, which will deposit on the counter electrode (lithium). The contents of transition metal on the cycled lithium electrode were detected by ICP-MS, which are presented in Table 3. It can be seen from Table 3 that more transition metals, especially nickel, have deposited on lithium foil electrode in baseline electrolyte than that in TMB-containing electrolyte, confirming the protection that cathode film provides.

4. Conclusion

The fully charged layered lithium-rich oxide (up to 4.8 V vs. Li/Li^+) suffers serious self-discharge in the carbonate-based electrolyte. This self-discharge results from the chemical interaction between the charged oxide and electrolyte, which involves the oxidation decomposition of the electrolyte and the structural destruction of the oxide. Trimethyl borate (TMB) is an effective electrolyte additive for suppressing this self-discharge. TMB is oxidized preferentially to the carbonate solvents in the electrolyte, forming a thin and protective cathode film on the oxide. With this cathode film, layered lithium-rich oxide is separated from direct contact with electrolyte, the electrolyte decomposition is suppressed and the structure integrity of the oxide is maintained.

Acknowledgments

This work is financially supported from the National Natural Science Foundation (Grant No. 21273084), the joint project of National Natural Science Foundation of China and Natural Science Foundation of Guangdong Province (Grant No. U1401248), the key projects of Science and Technology in Guangdong Province (Grant Nos. 2014B010123002 and 2015B010116002), the Innovative Platform of Guangzhou Municipality (Grant No. 201509030005), and the scientific research project of Department of Education of Guangdong Province (Grant No. 2013CXZDA013).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2016.05.074>.

References

- [1] X.D. Xiang, L.D. Xing, W.S. Li, *Sci. Adv. Mater* 6 (2014) 1506.
- [2] H.B. Lin, Y.M. Zhang, H.B. Rong, S.W. Mai, J.N. Hu, Y.H. Liao, L.D. Xing, M.Q. Xu, X.P. Li, W.S. Li, *J. Mater. Chem. A* 2 (2014) 11987.
- [3] X.L. Liao, Q.M. Huang, S.W. Mai, X.S. Wang, M.Q. Xu, L.D. Xing, Y.H. Liao, W.S. Li, *J. Power Sources* 272 (2014) 501.
- [4] J.M. Tarascon, M. Armand, *Nature* 414 (2001) 359.
- [5] C.R. Wu, X.P. Fang, X.W. Guo, Y. Mao, J. Ma, C.C. Zhao, Z.X. Wang, L.Q. Chen, *J. Power Sources* 231 (2013) 44.
- [6] L. Croguennec, J. Bains, M. Ménétrier, A. Flambard, E. Bekeart, C. Jordy, P. Biensan, C. Delmas, *J. Electrochem. Soc.* 156 (2009) A349.
- [7] S.H. Kang, M.M. Thackeray, *J. Electrochem. Soc.* 155 (2008) A269.
- [8] H. Yu, H. Zhou, *J. Phys. Chem. Lett.* 4 (2013) 1268.
- [9] A. Ito, Y. Sato, T. Sanada, M. Hatano, H. Horie, Y. Ohsawa, *J. Power Sources* 196 (2001) 6828.
- [10] J. Geder, J.H. Song, S.H. Kang, D.Y. Yu, *Solid State Ionics* 268 (2014) 242.
- [11] L.D. Xing, W. Li, C. Wang, M. Xu, X. Meng, S. Zhao, *J. Phys. Chem. B* 113 (2009) 5181.
- [12] L.D. Xing, W. Li, C. Wang, F.L. Gu, M. Xu, C.L. Tan, J. Yi, *J. Phys. Chem. B* 113 (2009) 16596.
- [13] A.D. Robertson, P.G. Bruce, *Chem. Mater* 15 (2003) 1984.
- [14] D.Y.W. Yu, K. Yanagida, Y. Kato, H. Nakamura, *J. Electrochem. Soc.* 156 (2009) A417.
- [15] S.H. Yu, T. Yoon, J. Mun, S. Park, Y.S. Kang, J.H. Park, S.M. Oh, Y.E. Sung, *J. Mater. Chem. A* 1 (2013) 2833.
- [16] J.G. Han, S.J. Lee, J. Lee, J.S. Kim, K.T. Lee, N.S. Choi, *Appl. Mater. Interfaces* 7 (2015) 8319.
- [17] C. Yu, G.S. Li, X.F. Guan, J. Zheng, D. Luo, L.P. Li, *Phys. Chem. Chem. Phys.* 14 (2012) 12368.
- [18] D.L. Ye, K. Ozawa, B. Wang, D. Hulicova-Jurcakova, J. Zou, C.H. Sun, L.Z. Wang, *Nano Energy* 6 (2014) 12368.
- [19] K. Xu, *Chem. Rev.* 104 (2004) 4303.
- [20] L. Yang, B. Ravel, B.L. Lucht, *Electrochem. Solid-State Lett.* 13 (2010) A95.
- [21] M.M. Thackeray, S.H. Kang, C.S. Johnson, J.T. Vaughey, R. Benedek, S.A. Hackney, *J. Mater. Chem.* 17 (2007) 3112.
- [22] B. Xu, C.R. Fell, M.F. Chi, Y.S. Meng, *Energy Environ. Sci.* 4 (2011) 2223.
- [23] H.J. Yu, H.J. Kim, Y.R. Wang, P. He, D. Asakura, Y. Nakamura, H.S. Zhou, *Phys. Chem. Chem. Phys.* 14 (2012) 6584.
- [24] A. Boulineau, L. Simonin, J.F. Colin, C. Bourbon, S. Patoux, *Nano Lett.* 13 (2013) 3857.
- [25] M. Gu, I. Belharouak, J.M. Zheng, H.M. Wu, J. Xiao, A. Genc, K. Amine, S. Thevuthasan, D.R. Baer, J.G. Zhang, N.D. Browning, J. Liu, C.M. Wang, *ACS Nano* 7 (2013) 760.
- [26] J.M. Zheng, M. Gu, J. Xiao, P.J. Zuo, C.M. Wang, J.G. Zhang, *Nano Lett.* 13 (2013) 3824.
- [27] P. Oh, M. Ko, S. Myeong, Y. Kim, J. Cho, *Adv. Energy Mater* 4 (2014) 1400631.
- [28] Z.H. Chen, Y.K. Sun, K. Amine, *J. Electrochem. Soc.* 153 (2006) A1818.
- [29] Y. Li, M. Bettge, B. Polzin, Y. Zhu, M. Balasubramanian, D.P. Abraham, *J. Electrochem. Soc.* 160 (2013) A3006.
- [30] Z.H. Lu, J.R. Dahn, *J. Electrochem. Soc.* 149 (2002) A815.
- [31] X.L. Liao, Q.M. Huang, S.W. Mai, X.S. Wang, M.Q. Xu, L.D. Xing, Y.H. Liao, W.S. Li, *J. Power Sources* 286 (2015) 551.
- [32] W. He, J. Qian, Y. Cao, X. Ai, H. Yang, *RSC Adv.* 2 (2012) 3423.
- [33] S.J. Shi, J.P. Tu, Y.Y. Tang, X.Y. Liu, Y.Q. Zhang, X.L. Wang, C.D. Gu, *Electrochim. Acta* 88 (2013) 671.
- [34] F. Wu, N. Li, Y. Su, H. Lu, L. Zhang, R. An, Z. Wang, L. Bao, S. Chen, *J. Mater. Chem.* 22 (2012) 1489.
- [35] Y. Wu, A. Murugan, A. Manthiram, *J. Electrochem. Soc.* 155 (2008) A635.
- [36] Q.Y. Wang, J. Liu, A.V. Murugan, A. Manthiram, *J. Mater. Chem.* 19 (2009) 4965.
- [37] Y.K. Sun, M.J. Lee, C.S. Yoon, J. Hassoun, K. Amine, B. Scrosati, *Adv. Mater* 24 (2012) 1192.
- [38] K.J. Rosina, M. Jiang, D. Zeng, E. Salager, A.S. Best, C.P. Grey, *J. Mater. Chem.* 22 (2012) 20602.
- [39] L.N. Cong, X.G. Gao, S.C. Ma, X. Guo, Y.P. Zeng, L.H. Tai, R.S. Wang, H.M. Xie, L.Q. Sun, *Electrochim. Acta* 115 (2014) 399.
- [40] J. Liu, A. Manthiram, *J. Mater. Chem.* 20 (2010) 3961.
- [41] Q.Y. Wang, J. Liu, A.V. Murugan, A. Manthiram, *J. Mater. Chem.* 19 (2009) 4965.
- [42] Y. Zhu, M. Casselman, Y. Li, A. Wei, D. Abraham, *J. Power Sources* 264 (2014) 184.
- [43] S. Tan, Z.R. Zhang, Y.X. Li, Y. Li, J.M. Zheng, Z.B. Zhou, Y. Yang, *J. Electrochem. Soc.* 160 (2013) A285.
- [44] A. Abouimrane, S.A. Odom, H. Tavassol, M.V. Schulmerich, H. Wu, R. Bhargava, A.A. Gewirth, J.S. Moore, K. Amine, *J. Electrochem. Soc.* 160 (2013) A268.
- [45] Y. Zhu, Y. Li, M. Bettge, D.P. Abraham, *J. Electrochem. Soc.* 159 (2012) A2109.
- [46] J. Zhang, J.L. Wang, J. Yang, Y.N. Li, *Electrochim. Acta* 117 (2014) 99.
- [47] J. Li, L. Xing, R. Zhang, M. Chen, Z. Wang, M. Xu, W. Li, *J. Power Sources* 285 (2015) 360.
- [48] Z.S. Wang, L.D. Xing, J.H. Li, B. Li, M.Q. Xu, Y.H. Liao, W.S. Li, *Electrochim. Acta* 184 (2015) 40.
- [49] Y. Han, K. Lee, J. Yoo, Y. Huh, *Theor. Chem. Acc.* 133 (2014) 1562.
- [50] Y. Qin, Z. Chen, H. Lee, X. Yang, K. Amine, *J. Phys. Chem. C* 144 (2010) 15202.
- [51] Y.B. Liu, T. Lei, L. Lei, *J. Power Sources* 221 (2013) 90.
- [52] H.B. Rong, M.Q. Xu, B.Y. Xie, W.Z. Huang, X.L. Liao, L.D. Xing, W.S. Li, *J. Power Sources* 274 (2015) 1155.
- [53] M. Chen, X. Xiang, D. Chen, Y. Liao, Q. Huang, W. Li, *J. Power Sources* 279 (2015) 197.
- [54] Z.L. Wang, N. Dupre, L. Lajaunie, P. Moreau, J. Martin, L. Boutafa, S. Patoux, D. Guyomard, *J. Power Source* 215 (2012) 170.
- [55] B. Li, Y.Q. Wang, W.Q. Tu, Z.S. Wang, M.Q. Xu, L.D. Xing, W.S. Li, *Electrochim. Acta* 273 (2014) 636.
- [56] S.W. Mai, M.Q. Xu, X.L. Liao, L.D. Xing, W.S. Li, *J. Power Sources* 273 (2015) 816.
- [57] X.D. Xiang, W.S. Li, *Electrochim. Acta* 133 (2014) 422.
- [58] T.A. Arinkumar, Y. Wu, A. Manthiram, *Chem. Mater* 19 (2007) 3067.
- [59] B.H. Song, M.O. Lai, L. Lu, *Electrochim. Acta* 80 (2012) 187.
- [60] B.H. Song, Z.W. Liu, M.O. Lai, L. Lu, *Phys. Chem. Chem. Phys.* 14 (2012) 12875.
- [61] D.S. Lu, W.S. Li, X.X. Zuo, Z.Z. Yuan, Q.M. Huang, *J. Phys. Chem. C* 111 (2007) 12067.

- [62] H.B. Rong, M.Q. Xu, B.Y. Xie, X.L. Liao, W.Z. Huang, L.D. Xing, W.S. Li, *Electrochim. Acta* 147 (2014) 31.
- [63] W.N. Huang, L.D. Xing, Y.T. Wang, M.Q. Xu, W.S. Li, F.C. Xie, S.G. Xia, *J. Power Source* 267 (2014) 560.
- [64] M. Xu, L. Hao, Y. Liu, W. Li, L. Xing, B. Li, *J. Phys. Chem. C* 115 (2011) 6085.
- [65] B. Liu, Y.Q. Wang, H.B. Lin, J.S. Liu, L.D. Xing, M.Q. Meng, W.S. Li, *Electrochim. Acta* 14 (2014) 263.
- [66] P. Verma, P. Maire, P. Novák, *Electrochim. Acta* 55 (2010) 6332.
- [67] M. Xu, A. Xiao, W. Li, B.L. Lucht, *J. Electrochem. Soc.* 157 (2010) A115.
- [68] D. Aurbach, I. Weissman, A. Schechter, H. Cohen, *Langmuir* 12 (1996) 3991.
- [69] W. Li, A. Xiao, B.L. Lucht, M.C. Smart, B.V. Ratnakumar, *J. Electrochem. Soc.* 155 (2008) A648.
- [70] L. Yang, T. Markmaitree, B.L. Lucht, *J. Power Sources* 196 (2011) 2251.
- [71] X. Wei, S. Zhang, Z. Du, P. Yang, J. Wang, Y. Ren, *Electrochim. Acta* 107 (2013) 549.